# Integrale cu parametrii. Functii $\Gamma$, $B$

1) $\int_0^\infty \frac{\sqrt[3]{x}}{(1+x^2)^3}\,dx \overset{\text{Not}}{=\!=} J$

Notăm $x^2 = y \Rightarrow x = \sqrt{y}$

$$dx = \frac{1}{2\sqrt{y}}\,dy$$

$$J = \int_0^\infty \frac{y^{1/6}}{(1+y)^3}\cdot\frac{1}{2\sqrt{y}}\,dy = \frac{1}{2}\int_0^\infty \frac{y^{-1/3}}{(1+y)^3}\,dy$$

$p-1 = -\frac{1}{3} \Rightarrow p = \frac{2}{3}$

$p+q = 3 \longrightarrow q = 3 - \frac{2}{3} = \frac{7}{3}$

$$\Rightarrow J = \frac{1}{2}B\left(\frac{2}{3};\frac{7}{3}\right)$$

$$J = \frac{\Gamma\left(\frac{2}{3}\right)\Gamma\left(\frac{7}{3}\right)}{2\cdot\Gamma(3)} = \frac{\Gamma\left(\frac{2}{3}\right)\Gamma\left(1+\frac{4}{3}\right)}{2\cdot\Gamma(3)} = \frac{\Gamma\left(\frac{2}{3}\right)\cdot\frac{4}{3}\Gamma\left(\frac{4}{3}\right)}{2\cdot 2!}$$

$$J = \frac{1}{3}\Gamma\left(\frac{2}{3}\right)\cdot\Gamma\left(1+\frac{1}{3}\right) = \frac{1}{3}\Gamma\left(\frac{2}{3}\right)\cdot\frac{1}{3}\Gamma\left(\frac{1}{3}\right) = \frac{1}{9}\cdot\frac{\pi}{\sin\frac{\pi}{3}} = \frac{2\pi}{9\sqrt{3}}$$

2) $\int_0^\infty x^4 e^{-x^2}\,dx = \frac{1}{2}\int_0^\infty t\sqrt{t}\,e^{-t}\,dt = \frac{1}{2}\int_0^\infty t^{3/2}e^{-t}\,dt$

Notăm $x^2 = t \Rightarrow x = \sqrt{t}$ $\quad x=0 \Rightarrow t=0$

$2x\,dx = dt$ $\quad x=\infty \Rightarrow t=\infty$

$$\alpha - 1 = \frac{3}{2} \Rightarrow \alpha = \frac{5}{2} \Rightarrow \int_0^\infty x^4 e^{-x^2}\,dx = \frac{1}{2}\Gamma\left(\frac{5}{2}\right)$$

$$\Gamma\left(\frac{5}{2}\right) = \Gamma\left(1+\frac{3}{2}\right) = \frac{3}{2}\Gamma\left(\frac{3}{2}\right) = \frac{3}{2}\Gamma\left(1+\frac{1}{2}\right) = \frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{3}{4}\Gamma\left(\frac{1}{2}\right)$$

$$\int_0^{\infty} x^4 e^{-x^2}\,dx = \frac{3}{8}\sqrt{\pi}$$

3) $$\int_0^1 \sqrt{\ln^3 \frac{1}{x}}\,dx$$

Notăm $\ln \frac{1}{x} = y \Rightarrow \frac{1}{x} = e^y \Rightarrow x = e^{-y}$

$x = 0 \Rightarrow y = \infty$ $\qquad dx = -e^{-y}\,dy$

$x = 1 \Rightarrow y = 0$

$$\int_0^1 \sqrt{\ln^3 \frac{1}{x}}\,dx = \int_{\infty}^0 y^{3/2}(-e^{-y})\,dy = \int_0^{\infty} y^{3/2} e^{-y}\,dy \Rightarrow$$

$\alpha - 1 = \frac{3}{2} \Rightarrow \alpha = \frac{5}{2}$

$$\Rightarrow \int_0^1 \sqrt{\ln^3 \frac{1}{x}}\,dx = \Gamma\left(\frac{5}{2}\right) = \Gamma\left(1+\frac{3}{2}\right) = \frac{3}{2}\Gamma\left(\frac{3}{2}\right) = \frac{3}{2}\Gamma\left(1+\frac{1}{2}\right)$$

$$= \frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{3}{4}\cdot\sqrt{\pi}$$

4) $$\int_0^1 \frac{x^m}{\sqrt{1-x^2}}\,dx \qquad m \in \mathbb{N}^*$$

Notăm $x^2 = t \Rightarrow x = \sqrt{t} \Rightarrow x^{m-1} = t^{\frac{m-1}{2}}$

$2x\,dx = dt$

$x = 0 \Rightarrow t = 0$

$x = 1 \Rightarrow t = 1$

$$\int_0^1 \frac{x^m}{\sqrt{1-x^2}}\,dx = \frac{1}{2}\int_0^1 t^{\frac{m-1}{2}}(1-t)^{-1/2}\,dt$$

$p - 1 = \frac{m-1}{2} \Rightarrow p = \frac{m-1}{2} + 1 = \frac{m+1}{2} \Rightarrow$

$q - 1 = -\frac{1}{2} \Rightarrow q = \frac{1}{2}$

$$\Rightarrow \int_0^1 \frac{x^m}{\sqrt{1-x^2}}\,dx = \frac{1}{2}B\left(\frac{m+1}{2},\frac{1}{2}\right) = \frac{1}{2}\cdot\frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(1+\frac{m}{2}\right)} =$$

$$= \frac{1}{2}\cdot\frac{\Gamma\left(1+\frac{m-1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(1+\frac{m}{2}\right)} = \frac{1}{2}\cdot\frac{\frac{m-1}{2}\Gamma\left(\frac{m-1}{2}\right)\sqrt{\pi}}{\frac{m}{2}\Gamma\left(\frac{m}{2}\right)} = \frac{(m-1)\sqrt{\pi}\,\Gamma\left(\frac{m-1}{2}\right)}{2m\,\Gamma\left(\frac{m}{2}\right)}$$

**5)** $\int_0^{\pi/2} \sin^{3/2}x\cos^{5/2}x\,dx \overset{\text{not. } y}{=\!=} y$

Notăm $\sin^2(x) = t \longrightarrow$ $x=0 \Rightarrow t=0$

$2\sin x\cos x = dt$ $\quad x=\frac{\pi}{2} \Rightarrow t=1$

$$y = \frac{1}{2}\int_0^1 \underbrace{\sin^{1/2}x}_{(\sin^2x)^{1/4}}\cdot\underbrace{\cos^{3/2}x}_{(\cos^2x)^{3/4}}\cdot\underbrace{2\cdot\sin x\cos x\,dx}_{dt}$$

$(\cos^2x)^{3/4} = (1-\sin^2x)^{3/4}$

$$y = \frac{1}{2}\int_0^1 t^{1/4}(1-t)^{3/4}\,dt$$

$p-1=\frac{1}{4} \Rightarrow p=\frac{5}{4}$

$q-1=\frac{3}{4} \Rightarrow q=\frac{7}{4}$

$$\Rightarrow y = \frac{1}{2}B\left(\frac{5}{4};\frac{7}{4}\right) = \frac{1}{2}\cdot\frac{\Gamma\left(\frac{5}{4}\right)\Gamma\left(\frac{7}{4}\right)}{\Gamma(3)}$$

$$y = \frac{1}{2}\cdot\frac{\frac{1}{4}\Gamma\left(\frac{1}{4}\right)\cdot\frac{3}{4}\Gamma\left(\frac{3}{4}\right)}{3!} \Rightarrow$$

$$\Rightarrow y = \frac{\Gamma\left(\frac{1}{4}\right)\cdot\Gamma\left(\frac{3}{4}\right)}{16\cdot 2\cdot 2} = \frac{1}{64}\cdot\frac{\pi}{\sin\frac{\pi}{4}} = \frac{1}{64}\cdot\frac{\pi\cdot 2\sqrt{2}}{\cancel{2}} = \frac{\pi\sqrt{2}}{64}$$

**6)** $\int_{-\infty}^{\infty} e^{-2x^2+x+1}\,dx = \int_{-\infty}^{\infty} e^{-2\left(x^2-\frac{1}{2}x-\frac{1}{2}\right)}\,dx =$

$$= \int_{-\infty}^{\infty} e^{-2\left[(x-1/4)^2 - 9/16\right]}\,dx = e^{9/8}\int_{-\infty}^{\infty} e^{-2(x-1/4)^2}\,dx \overset{\text{Not}}{=\!=} I$$

Not. $\sqrt{2}(x-1/4) = y$

$\sqrt{2}\,dx = dy$

$$\int_{-\infty}^{\infty} e^{-y^2} = \sqrt{\pi}$$

$$I = e^{9/8}\int_{-\infty}^{\infty} e^{-y^2}\cdot\frac{1}{\sqrt{2}}\,dy = \frac{e^{9/8}\cdot\sqrt{\pi}}{\sqrt{2}}$$

7) $$\int_{-4}^{3} \frac{dx}{\sqrt[6]{(x+4)(x-3)^5}} \overset{\text{Notăm } y}{=\!=}$$

Notăm $\frac{x+4}{7} = y \longrightarrow$ $x = -4 \Rightarrow y = 0$

$x = 3 \Rightarrow y = 1$

$\frac{dx}{7} = dy \Rightarrow dx = 7\,dy$

$$I = \int_0^1 \frac{7\,dy}{\sqrt[6]{7y(7y-7)^5}} = \int_0^1 \frac{dy}{y^{1/6}(y-1)^{5/6}} = \int_0^1 y^{-1/6}(1-y)^{-5/6}\,dy$$

$p-1 = -\frac{1}{6} \Rightarrow p = \frac{5}{6}$

$q-1 = -\frac{5}{6} \Rightarrow q = \frac{1}{6}$

$$\Rightarrow I = B(5/6, 1/6) = \frac{\Gamma(5/6)\,\Gamma(1/6)}{\Gamma(1)}$$

$$I = \frac{\pi}{\sin\frac{\pi}{6}} = 2\pi$$

8) $$\int_0^\infty \frac{x\sqrt{x}}{(x^2-2x+4)^2}\,dx = \int_0^\infty \frac{x^{3/2}(x+2)^2}{(x^3+8)^2}\,dx = \int_0^\infty \frac{x^{7/2}+4x^{5/2}+4x^{3/2}}{(x^3+8)^2}\,dx$$

$$= \underbrace{\int_0^\infty \frac{x^{7/2}}{(x^3+8)^2}\,dx}_{J_1} + 4\underbrace{\int_0^\infty \frac{x^{5/2}}{(x^3+8)^2}\,dx}_{J_2} + 4\underbrace{\int_0^\infty \frac{x^{3/2}}{(x^3+8)^2}\,dx}_{J_3}$$

Notăm $x^3 = y \Rightarrow x = \sqrt[3]{y}$

$$dx = \frac{1}{3\sqrt[3]{y^2}}\,dy$$

$$J_1 = \int_0^\infty \frac{y^{7/6}}{(y+8)^2}\cdot\frac{1}{3\sqrt[3]{y^2}}\,dy = \frac{1}{3}\int_0^\infty \frac{y^{1/2}}{(y+8)^2}\,dy = \frac{1}{3}\int_0^\infty \frac{y^{1/2}\,dy}{\left[8\left(\frac{y}{8}+1\right)\right]^2}$$

Notăm $\frac{y}{8} = t \Rightarrow y = 8t$

$dy = 8dt$ $\quad J_1 = \frac{1}{3}\cdot\frac{1}{64}\cdot\int_0^\infty \frac{(8t)^{1/2}}{(t+1)^2}\,8\,dt$

$$J_1 = \frac{1}{3\sqrt{8}}\int_0^\infty \frac{t^{1/2}}{(t+1)^2}\,dt$$

$p-1=1/2 \Rightarrow p=3/2$ $\quad p+q=2 \Rightarrow q=1/2$

$$\Rightarrow J_1 = \frac{1}{3\sqrt{8}}B(3/2,1/2)$$

$$J_1 = \frac{1}{3\sqrt{8}}\cdot\frac{\Gamma(3/2)\Gamma(1/2)}{\Gamma(2)} \Rightarrow$$

$$\Gamma(3/2) = \Gamma(1+1/2) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}$$

$$\Rightarrow J_1 = \frac{1}{3\sqrt{8}}\,\frac{\frac{\sqrt{\pi}}{2}\cdot\sqrt{\pi}}{2} = \frac{\pi}{12\sqrt{8}} = \frac{\pi}{24\sqrt{2}}$$

$$I_2=\int_0^{\infty}\frac{x^{5/2}}{(x^3+8)^2}dx=\int_0^{\infty}\frac{x^{5/2}}{\left[8\left(\frac{x^3}{8}+1\right)\right]^2}dx$$

Notăm $\frac{x^3}{8}=y \Rightarrow x^3=8y \Rightarrow x=2\sqrt[3]{y}$

$$dx=2\cdot\frac{1}{3\sqrt[3]{y^2}}dy$$

$$I_2=\frac{1}{8^2}\int_0^{\infty}\frac{\sqrt{2^5}\cdot\sqrt[6]{y^5}}{(y+1)^2}\cdot 2\cdot\frac{dy}{3\sqrt[3]{y^2}}=\frac{2^3\cdot\sqrt{2}}{64\,3}\int_0^{\infty}\frac{y^{\frac{5}{6}}\cdot y^{-\frac{2}{3}}}{(y+1)^2}dy$$

$$I_2=\frac{\sqrt{2}}{24}\int_0^{\infty}\frac{y^{1/6}}{(y+1)^2}dy \Rightarrow I_2=\frac{\sqrt{2}}{24}B(7/6,5/6)$$

$p-1=1/6 \Rightarrow p=7/6$

$p+q=2 \Rightarrow q=\frac{5}{6}$

$$I_2=\frac{\sqrt{2}}{24}\cdot\frac{\Gamma(7/6)\Gamma(5/6)}{\Gamma(2)}$$

$\Gamma(7/6)=\Gamma(1+1/6)=1/6\,\Gamma(1/6)$

$$\Gamma(1/6)\Gamma(5/6)=\frac{\pi}{\sin\frac{\pi}{6}}=2\pi$$

$$\Rightarrow I_2=\frac{\sqrt{2}}{24}\cdot\frac{\Gamma(1/6)\cdot 1/6\cdot\Gamma(5/6)}{1!}=\frac{\sqrt{2}}{24}\cdot\frac{2\pi\cdot 1}{6}=\frac{\pi\sqrt{2}}{72}$$

$$I_3=\int_0^{\infty}\frac{x^{3/2}}{(x^3+8)^2}dx=\int_0^{\infty}\frac{x^{3/2}}{\left[8\left(\frac{x^3}{8}+1\right)\right]^2}dx \Rightarrow I_3=\frac{2}{3\cdot 64}\int_0^{\infty}\frac{\sqrt{2^3}\cdot\sqrt{y}}{(y+1)^2}\cdot\frac{dy}{\sqrt[3]{y^2}}$$

$$\frac{2}{3\cdot 64}=\frac{\sqrt{2}}{3\cdot 32}=\frac{\sqrt{2}}{96}$$

Notăm $\frac{x^3}{8}=y \Rightarrow x^3=8y \Rightarrow x=2\sqrt[3]{y}$

$$dx=2\cdot\frac{1}{3\sqrt[3]{y^2}}dy$$

$$I_3 = \frac{\sqrt{2}}{3\cdot 16}\int_0^{\infty} \frac{y^{1/2}\cdot y^{-2/3}}{(y+1)^2}\,dy = \frac{\sqrt{2}}{3\cdot 16}\cdot\int_0^{\infty}\frac{y^{-1/6}}{(y+1)^2}\,dy$$

$$p-1=-1/6 \Rightarrow p=5/6 \quad p+q=2 \Rightarrow q=\frac{12-5}{6}=\frac{7}{6} \Rightarrow I_3=\frac{\sqrt{2}}{3\cdot 16}B\left(\frac{5}{6};\frac{7}{6}\right)$$

$$I_3=\frac{\sqrt{2}}{3\cdot 16}\cdot\frac{\Gamma(5/6)\cdot\Gamma(7/6)}{\Gamma(2)}=\frac{\sqrt{2}}{3\cdot 16}\cdot\frac{\Gamma(5/6)\,\Gamma(1+1/6)}{\Gamma(2)}$$

$$=\frac{\sqrt{2}}{3\cdot 16}\cdot\frac{\Gamma(5/6)\,\frac{1}{6}\,\Gamma(1/6)}{1!}=\frac{\sqrt{2}}{3\cdot 16}\cdot\frac{2\pi}{6}=\frac{\pi\sqrt{2}}{144}$$

$$I=I_1+4I_2+4I_3=\frac{\pi}{12\sqrt{2}}+4\cdot\frac{5\pi\sqrt{2}}{72}+4\cdot\frac{\pi\sqrt{2}}{144}$$

$$I=\frac{3}{\pi\sqrt{2}}{24}+\frac{4}{5\pi\sqrt{2}}{18}+\frac{2}{\pi\sqrt{2}}{36}=\frac{\pi\sqrt{2}(3+20+2)}{72}=\frac{25\pi\sqrt{2}}{72}$$

9) $$I(a)=\int_0^{\pi/2}\ln\frac{1+a\sin x}{1-a\sin x}\,\frac{dx}{\sin x} \qquad |a|<1$$

$$I'(a)=\int_0^{\pi/2}\frac{1-a\sin x}{1+a\sin x}\cdot\frac{\sin x(1-a\sin x)+(1+a\sin x)\sin x}{(1-a\sin x)^2}\,\frac{dx}{\sin x}$$

$$I'(a)=\int_0^{\pi/2}\frac{\sin x(1-a\sin x+1+a\sin x)}{(1+a\sin x)(1-a\sin x)}\cdot\frac{dx}{\sin x}=2\int_0^{\pi/2}\frac{1}{1-a^2\sin^2 x}\,dx$$

Notăm $\operatorname{tg} x = t \Rightarrow x=\operatorname{arctg} t$

$$dx=\frac{1}{1+t^2}\,dt \Rightarrow$$

$$t^2=\operatorname{tg}^2 x=\frac{\sin^2 x}{\cos^2 x}=\frac{\sin^2 x}{1-\sin^2 x} \Rightarrow t^2-t^2\sin^2 x=\sin^2 x$$

$$\Rightarrow \sin^2 x = \frac{t^2}{1+t^2}$$

$$\Rightarrow J'(a) = 2\int_0^\infty \frac{1}{1-a^2\frac{t^2}{1+t^2}} \cdot \frac{1}{1+t^2}\,dt = 2\int_0^\infty \frac{1}{1+t^2-a^2t^2}\,dt$$

$$J'(a) = 2\int_0^\infty \frac{1}{1+(1-a^2)t^2}\,dt = \frac{2}{1-a^2}\int_0^\infty \frac{1}{\frac{1}{1-a^2}+t^2}\,dt$$

$$J'(a) = \frac{2}{1-a^2}\cdot\sqrt{1-a^2}\cdot \operatorname{arctg} t\sqrt{1-a^2}\Big|_0^\infty$$

$$J'(a) = \frac{2}{\sqrt{1-a^2}}\cdot\left[\frac{\pi}{2}-0\right] = \frac{\pi}{\sqrt{1-a^2}}$$

$$J(a) = \int \frac{\pi}{\sqrt{1-a^2}}\,da = \pi \arcsin a + C$$

$$J(0) = \pi \arcsin 0 + C = C$$

$$J(0) = 0$$

$$\Rightarrow C = 0 \qquad \Rightarrow J(a) = \pi \arcsin a$$